

Practice set 1: Discrete Mathematics (CSE121)

Instruction. This document is solely for practice. Do not submit the answers. Also, the answer keys to these questions will not be provided.

1 Problem set

Question 1.1. Let p, q, r be propositions. Write the truth tables of the following.

1. $p \implies q \implies r \implies p$.
2. $p \iff q \iff (r \oplus \neg r)$.
3. $p \wedge q \vee r \wedge \neg p \vee \neg q \wedge \neg r$.
4. $p \iff (q \wedge \neg r) \implies (q \iff r) \wedge (\neg p \oplus \neg q)$.
5. $r \implies q \implies p \iff \neg p$.
6. $(p \wedge q \vee \neg q) \oplus (p \vee \neg q \implies r)$.
7. $\neg(p \wedge q \neg r) \iff \neg(r \implies \neg q)$.
8. $\neg(p \wedge q \iff \neg r) \oplus \neg(\neg r \vee q \implies p)$.
9. $\neg p \wedge \neg q \vee \neg r \implies \neg(p \vee q \vee r)$.
10. $(p \oplus \neg q) \wedge (q \implies \neg r) \wedge (r \iff \neg p)$,

Question 1.2. 1. Let p, q, r be propositions. Write CNFs and DNFs equivalent to the following propositions using the truth table approach.

- (a) $p \implies (q \oplus \neg r)$.
 - (b) $(p \oplus q) \iff \neg(p \wedge \neg r)$.
 - (c) $(p \implies \neg q) \wedge (q \iff \neg r)$.
 - (d) $\neg(p \wedge q) \vee \neg(q \vee r)$.
 - (e) $\neg(p \wedge q) \implies \neg(\neg q \vee r)$.
2. Let p, q, r be propositions. Write CNFs and DNFs equivalent to the following propositions using only the algebra of propositional logic. Verify if your answer matches the answer of the first.

- (a) $p \implies (q \oplus \neg r)$.
- (b) $(p \oplus q) \iff \neg(p \wedge \neg r)$.
- (c) $(p \implies \neg q) \wedge (q \iff \neg r)$.
- (d) $\neg(p \wedge q) \vee \neg(q \vee r)$.
- (e) $\neg(p \wedge q) \implies \neg(\neg q \vee r)$.

Question 1.3. *Convert the following statements from English to statements from predicate logic.*

1. *Every student likes mathematics.*
2. *Some cats are black.*
3. *There exists a number greater than 100.*
4. *Not every bird can fly.*
5. *No person lives forever.*
6. *Every city has a mayor.*
7. *Some animals are dangerous.*
8. *There exists a student who studies hard.*
9. *Not every teacher is strict.*
10. *No even number is odd.*
11. *Every parent loves their child.*
12. *Some students play football.*
13. *There exists a prime number less than 10.*
14. *Not every country is rich.*
15. *No student failed the exam.*
16. *Every book has some author.*
17. *Some flowers bloom at night.*
18. *There exists a teacher who is admired by all students.*
19. *Not every river flows into the sea.*
20. *No triangle has four sides.*
21. *Every person has some friend.*
22. *Some person is friends with everyone.*
23. *There exists a student who likes every subject.*

24. *Not every student likes every teacher.*
25. *No one likes every single person.*
26. *Every dog has some owner.*
27. *Some owner loves every dog they own.*
28. *There exists a book that every student has read.*
29. *Not every book is read by some student.*
30. *No student has read every book in the library.*
31. *Every teacher teaches some subject.*
32. *Some subjects are taught by every teacher.*
33. *There exists a subject that no teacher teaches.*
34. *Not every subject is interesting.*
35. *No subject is liked by every student.*
36. *Every house has some door.*
37. *Some houses have every modern facility.*
38. *There exists a city where every house has electricity.*
39. *Not every city has clean water.*
40. *No house has infinite rooms.*
41. *Every soldier serves some country.*
42. *Some countries are served by every soldier.*
43. *There exists a country that has no soldiers.*
44. *Not every soldier is brave.*
45. *No army is invincible.*
46. *Every company employs some worker.*
47. *Some companies employ every skilled worker.*
48. *There exists a company where every worker is happy.*
49. *Not every company is profitable.*
50. *No company lasts forever.*
51. *Every mathematician knows some theorem.*

- 52. *Some theorems are known by every mathematician.*
- 53. *There exists a theorem unknown to every mathematician.*
- 54. *Not every proof is short.*
- 55. *No theorem has infinitely many shortest proofs.*
- 56. *Every child has some toy.*
- 57. *Some toys are liked by every child.*
- 58. *There exists a toy liked by no child.*
- 59. *Not every toy is safe.*
- 60. *No toy lasts forever.*
- 61. *Every problem has some solution.*
- 62. *Some solutions solve every problem.*
- 63. *There exists a problem with no solution.*
- 64. *Not every problem is difficult.*
- 65. *No solution works for all problems.*
- 66. *Every student writes some exam.*
- 67. *Some exams are written by every student.*
- 68. *There exists an exam that no student writes.*
- 69. *Not every exam is fair.*
- 70. *No exam is passed by every student.*
- 71. *Every scientist works on some project.*
- 72. *Some projects are worked on by every scientist.*
- 73. *There exists a project that no scientist works on.*
- 74. *Not every project is funded.*
- 75. *No project succeeds without effort.*
- 76. *Every person believes some idea.*
- 77. *Some ideas are believed by everyone.*
- 78. *There exists an idea believed by no one.*
- 79. *Not every belief is true.*

80. *No false idea is believed by every person.*
81. *Every politician represents some citizen.*
82. *Some citizens are represented by every politician.*
83. *There exists a citizen represented by no politician.*
84. *Not every politician is honest.*
85. *No politician pleases every citizen.*
86. *Every worker has some task.*
87. *Some tasks are done by every worker.*
88. *There exists a task that no worker does.*
89. *Not every task is important.*
90. *No task is done by infinitely many workers.*
91. *Every course has some student enrolled.*
92. *Some students are enrolled in every course.*
93. *There exists a course with no students.*
94. *Not every student passes every course.*
95. *No student takes infinitely many courses.*
96. *Every machine performs some function.*
97. *Some functions are performed by every machine.*
98. *There exists a machine that performs no function.*
99. *Not every machine is reliable.*
100. *No machine works forever.*
101. *Every student in some class admires every teacher of that class.*
102. *There exists a class in which every student admires some teacher.*
103. *For every student, there exists some course such that the student dislikes every teacher of that course.*
104. *There exists a teacher admired by some student in every class.*
105. *Every person has some friend who knows every other person.*
106. *There exists a person who is a friend of every friend of some person.*

107. *Every child has some toy that no other child likes.*
108. *There exists a toy that is liked by some child in every family.*
109. *For every book, there exists some student who has read every chapter of that book.*
110. *There exists a student who has read some chapter in every book.*
111. *Every university has some department that admits every student.*
112. *There exists a university where every department admits some student.*
113. *For every doctor, there exists some patient who visits every clinic the doctor works at.*
114. *There exists a patient who visits some clinic of every doctor.*
115. *Every person has some hobby that no one else shares.*
116. *There exists a hobby that every person shares.*
117. *For every language, there exists some word understood by every speaker of that language.*
118. *There exists a word that is understood by some speaker in every language.*
119. *Every number has some multiple that is divisible by every prime.*
120. *There exists a number divisible by every prime.*
121. *For every teacher, there exists some student who asks every question in class.*
122. *There exists a student who asks some question to every teacher.*
123. *Every country has some law that applies to every citizen.*
124. *There exists a law that applies to some citizen in every country.*
125. *For every problem, there exists some solution that is checked by every examiner.*
126. *There exists a solution that solves some problem in every subject.*
127. *Every actor has some movie that is watched by every critic.*
128. *There exists a movie that is watched by some critic in every country.*
129. *For every city, there exists some road that connects it to every other city.*
130. *There exists a city connected by some road to every other city.*

Question 1.4. *First convert the following statements to predicate logic statements, then apply DeMorgan's rules of quantifiers and other rules of algebra of propositional logic to simplify these logical statements, and then convert these simplified logical statements into English statements.*

1. *Not every student likes mathematics.*
2. *Not all birds can fly.*

3. *No triangle has four sides.*
4. *No cat is green.*
5. *Not every person is tall.*
6. *No child is 200 years old.*
7. *Not all flowers bloom in winter.*
8. *There does not exist a number smaller than zero and larger than zero.*
9. *No book is infinite in length.*
10. *Not every car is expensive.*
11. *Not all teachers give homework.*
12. *No student solved every problem on the test.*
13. *Not every country is wealthy.*
14. *There does not exist a perfect square that is prime.*
15. *No musician can play every instrument.*
16. *Not every parent loves mathematics.*
17. *No person has visited every city in the world.*
18. *Not every animal lives on land.*
19. *There does not exist a library containing every book.*
20. *No day has more than 24 hours.*
21. *Not every number is rational.*
22. *No planet is closer to the Sun than Mercury.*
23. *Not all rivers reach the ocean.*
24. *There does not exist a student who never made a mistake.*
25. *No square is a circle.*
26. *Not every problem has an easy solution.*
27. *No person knows all the secrets of the universe.*
28. *Not every bird migrates.*
29. *There does not exist a natural number greater than all natural numbers.*
30. *No athlete wins every competition.*

31. *Not every theorem can be proved.*
32. *No machine solves every problem.*
33. *Not all languages are spoken by people today.*
34. *There does not exist a set containing all sets.*
35. *No infinite sequence contains every possible number.*
36. *Not every prime is even.*
37. *No computer can run forever without energy.*
38. *Not every question has an answer.*
39. *There does not exist a person who is admired by everyone.*
40. *No rule applies to every possible situation.*
41. *Not every statement is provable in mathematics.*
42. *No logical system captures all mathematical truths.*
43. *Not all problems are computable.*
44. *There does not exist a number that is both prime and divisible by every number.*
45. *No infinite set is smaller than every finite set.*
46. *Not every function is continuous.*
47. *No structure models all consistent theories.*
48. *Not every infinite series converges.*
49. *There does not exist a world where all contradictions are true.*
50. *No axiom system proves its own consistency.*